Entropy-Bounded Empiricism: Core Principles

A Regime-Based Epistemic Framework for Measurement Interpretation in Emergent Systems

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Abstract

This document presents the core principles of *Entropy-Bounded Empiricism* (EBE), a regime-based epistemic framework for interpreting measurements in systems governed by entropy gradients and energy flow. EBE introduces two orthogonal classification axes: the *S-axis* (physical regime hierarchy based on entropy state) and the *L-axis* (epistemic layer hierarchy from raw data to theoretical constructs). The *Regime-Consistent Measurement Protocol* (RCMP) defines valid mappings between these axes, preventing interpretive overreach and model misapplication. Although developed within the Energy-Flow Cosmology (EFC) framework, EBE generalizes to any domain where emergent phenomena require regime-aware interpretation—including biological, cognitive, ecological, economic, and technological systems. This document serves as the foundational specification for EBE/EFC research methodology.

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1 Introduction

Scientific models succeed or fail within specific *regimes*—domains where their primary variables directly track the physical drivers of observed phenomena. When models are applied outside these regimes, interpretive errors accumulate. Measurements become proxies for proxies, and theoretical constructs detach from empirical grounding.

EBE addresses this problem by making regime boundaries *explicit* and by defining clear rules for when measurements can validly inform theory. The framework does not replace existing physical models; rather, it provides a meta-level architecture that specifies:

(i) Which physical quantities are *driver-near* in each regime

(ii) How epistemic claims become progressively model-dependent

(iii) When interpretation remains valid versus when it becomes extrapolation

The result is not a new physics but a new *epistemology for physics*—and for any science dealing with emergent, entropy-driven systems.

2 Purpose and Scope

Definition 2.1 (Entropy-Bounded Empiricism)**.** EBE is a regime-based epistemic framework for interpreting measurements in systems where entropy gradients (∇S) govern emergent structure and dynamics.

EBE is:

- A *methodology*, not a physical model
- *Framework-independent*—applicable beyond EFC
- Designed to prevent *regime leakage* (applying models outside their validity domain)

The core insight is simple: *what we can know depends on where we measure.*

3 The S-Axis: Physical Regime Hierarchy

The S-axis classifies physical regimes by their dominant entropy character. The normalized entropy state $S \in [0, 1]$ determines which physical descriptions are appropriate.

Definition 3.1 (S-Axis Regime Classification)**.** The S-axis partitions physical phenomena into regimes based on local entropy state:

$$\mathcal{S} : S \in [0, 1] \longrightarrow \{\text{Low-}S, \text{Mid-}S, \text{High-}S\} \tag{1}$$

Table 1: S-Axis Regime Characteristics

Regime	**Physical Character**	**Dominant Description**
Low-S	Field/statistical ensemble	Spectral analysis, field theory
Mid-S	Structure/relational dynamics	Response functions, correlations
High-S	Local complexity/thermodynamic	Energy flows, entropy production

Principle 3.1 (Regime-Dependent Observables)**.** Which physical quantities are *meaningful* depends on the S-level of the phenomenon. A quantity that is a primary driver in one regime may be a derived proxy in another.

Remark 3.1. The S-axis is not merely a classification convenience. It reflects the physical fact that entropy gradients (∇S) change the causal structure of systems—what drives what depends on where you are in entropy space.

4 The L-Axis: Epistemic Layer Hierarchy

The L-axis classifies claims by their epistemic strength—how directly they connect to observation versus how much they depend on theoretical assumptions.

Definition 4.1 (L-Axis Epistemic Layers)**.**

$$\mathcal{L}_0 : \text{Raw data (direct observation)} \tag{2}$$
$$\mathcal{L}_1 : \text{Calibrated observables (instrument-corrected)} \tag{3}$$
$$\mathcal{L}_2 : \text{Derived physical quantities (regime-near physics)} \tag{4}$$
$$\mathcal{L}_3 : \text{Theoretical constructs (model-dependent interpretation)} \tag{5}$$

Table 2: L-Axis Layer Definitions

Layer	Type	Role
L_0	Raw data	Direct instrumental output; minimal assumptions
L_1	Calibrated observables	Corrected for known systematics; instrument models applied
L_2	Derived quantities	Physical parameters computed from L_1; regime-specific
L_3	Theoretical constructs	Model-dependent interpretation; highest assumption load

Principle 4.1 (Epistemic Degradation)**.** The higher the L-level, the more model-dependent the claim. Every step up the L-axis adds assumptions, increases uncertainty, and introduces regime-dependence.

Axiom 4.1 (Layer Transparency)**.** All interpretive chains must be explicitly traceable:

$$L_0 \xrightarrow{\text{calibration}} L_1 \xrightarrow{\text{derivation}} L_2 \xrightarrow{\text{modeling}} L_3 \tag{6}$$

Each arrow represents additional assumptions that must be documented.

5 RCMP: The Regime-Consistent Measurement Protocol

The RCMP is the central engine of EBE. It defines valid mappings between the physical regime (S-axis) and the epistemic layer (L-axis).

Definition 5.1 (Regime-Consistent Measurement Protocol)**.** For a phenomenon in regime S, interpretation must be anchored in the lowest L-level that remains *directly coupled* to the physical driver in that regime.

Principle 5.1 (RCMP Coupling Rule)**.**

$$\text{RCMP}(S) = \min\{L : L \text{ is driver-near in regime } S\} \tag{7}$$

This yields the following validity mapping:

Table 3: RCMP Regime-Layer Validity

S-Regime	Valid Primary L-Level
Field regime (Low-S)	L_1–L_2
Structure regime (Mid-S)	L_2
Complexity regime (High-S)	L_1–L_2 (locally)

Remark 5.1. L_3 claims are always permissible but must *never be treated as measurements.* They are interpretations, not data.

6 The Regime-Gating Principle

This is the core epistemological constraint of EBE:

Principle 6.1 (Regime Gating)**.** A model is *epistemically valid* only within the regime where its primary variables are driver-near.

$$\text{Validity}(M) = \{S : \text{primary variables of } M \text{ are driver-near in } S\} \tag{8}$$

This principle explains apparent "conflicts" between models:

- Newtonian mechanics is valid in the mid-S everyday regime
- General Relativity is valid in the low-S cosmological regime
- RAR (Radial Acceleration Relation) may be more fundamental than halo profiles in the galactic structure regime

Remark 6.1. No model is universal; no model is "wrong" within its regime. Regime Gating transforms model conflicts into regime boundary questions.

7 Proxy-Chain Transparency

All interpretations in EBE must show their derivation chain:

$$\underbrace{D}_{\text{Measurement } (L_0)} \xrightarrow{A_1} \underbrace{O}_{L_1} \xrightarrow{A_2} \underbrace{Q}_{L_2} \xrightarrow{A_3} \underbrace{T}_{L_3} \tag{9}$$

where:

- D = raw data
- O = calibrated observable
- Q = derived physical quantity
- T = theoretical construct
- A_i = assumption set at each transition

Principle 7.1 (Proxy-Chain Documentation)**.** Every claim must specify:

(a) Its L-level

(b) The proxy chain from L_0

(c) The assumptions at each step

(d) The S-regime where the chain is valid

This is EBE's version of scientific traceability—but with explicit regime awareness.

8 Epistemic Consequences

EBE leads to a crucial meta-scientific insight:

Principle 8.1 (Conflict Resolution)**.** Apparent conflicts between models often indicate that they operate at different S-levels or different L-levels—not that one is necessarily wrong.

This gives "everyone has part of the truth" a precise scientific form:

- Model A at S_1, L_2 and Model B at S_2, L_3 are not in conflict—they make claims in different epistemic coordinates
- True conflict only occurs when two models make incompatible claims at the *same* (S, L) coordinate

9 Summary: The EBE Core Equation

The entire EBE framework can be summarized in one statement:

EBE Core Statement

Physical validity depends on regime ($\mathcal{S}$), epistemic strength depends on layer ($\mathcal{L}$), and correct interpretation requires regime-consistent mapping between them (RCMP).

Or formally:

$$\text{Valid Interpretation} \iff \text{RCMP}(\mathcal{S}) \leq \mathcal{L} \leq L_3 \tag{10}$$

where $\text{RCMP}(\mathcal{S})$ returns the minimum valid L-level for regime $\mathcal{S}$.

10 Applications Beyond Cosmology

Although EBE was developed for cosmological modeling within the EFC framework, its principles generalize to any system governed by:

- Energy flow and entropy gradients
- Emergent structure at multiple scales
- Regime-dependent dynamics

Candidate domains include:

Table 4: EBE Applicability Across Domains

Domain	**EBE Application**
Biology	Metabolic regimes, morphogenesis
Neuroscience	Neural dynamics, consciousness studies
Ecology	Ecosystem stability, trophic cascades
Economics	Market regimes, complexity economics
Technology	AI systems, network dynamics

11 Conclusion

EBE provides an epistemic architecture specification: rules for determining what can be said, when, and at what level of confidence. It does not replace physical models but provides the meta-framework within which models can be properly compared, validated, and bounded.

The three pillars of EBE are:

1. **S-Axis**: Physical regime classification by entropy state
2. **L-Axis**: Epistemic layer classification by assumption load
3. **RCMP**: Valid mapping rules between regimes and layers

Together, they constitute a framework for *principled interpretation*—ensuring that empirical claims remain bounded by the regimes where they are epistemically valid.

Related Publications

[1] Magnusson, M. (2025). *Energy-Flow Cosmology: A Thermodynamic Bridge Between General Relativity and Quantum Field Theory.* DOI: 10.6084/m9.figshare.27315821

[2] Magnusson, M. (2026). *L0–L3 Regime Architecture in Entropy-Bounded Empiricism.* DOI: 10.6084/m9.figshare.31112536

Version History

Version	**Date**	**Changes**
1.0	January 2026	Initial release